Static Bench — Field Setup Sheet

DWG paper4-rung1 · rev A

Paper 4 · Rung 1, the atomic test: does a wall echo appear in ESP32 CSI? Two boards, one metal plate, a straight corridor. Everything measured with a tape.

METHOD	BANDWIDTH	RANGE CELL	GROUND TRUTH
Monostatic-in-effect bistatic RCS w/ bkgnd sub.	40 MHz (HT40)	3.75 m (c/2B)	Tape measure
TX BOARD	**RX BOARD**	**BASELINE B**	**PLATE D (START)**
28:84:85:48:40:20	28:84:85:53:99:DC	0.50 m	12.0 m

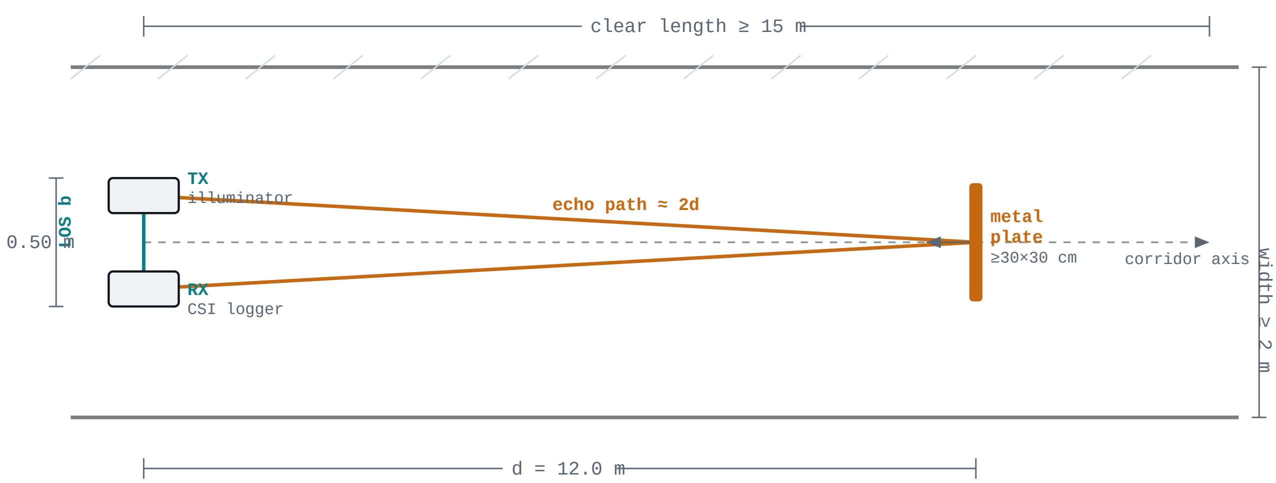
DRAWING 1 OF 4 — PLAN

A Plan view — looking down from above

The two boards sit together at the near end, **0.50 m** apart across the corridor. The plate stands on the corridor axis **12.0 m** away, facing the boards. Length is drawn to scale; the cross-corridor spacing is exaggerated so the two boards and their signal paths are legible.

Top-down geometry

scale: length 1:~ · width n.t.s.



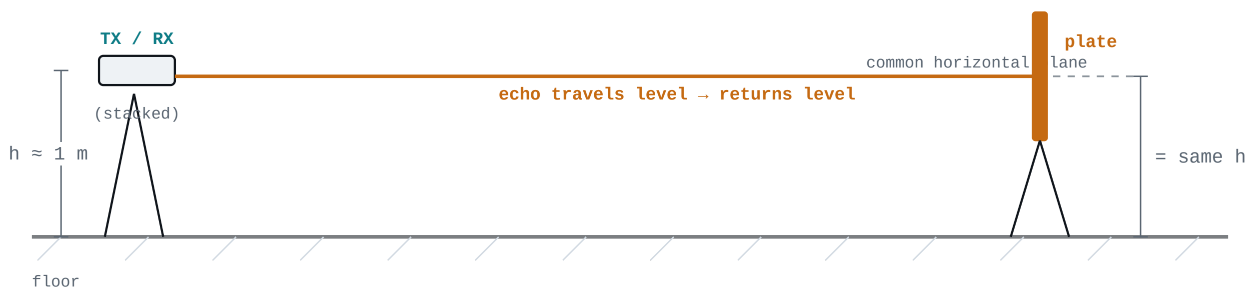
The 2.4° offset from the plate to each board (0.25 m over 12 m) is negligible — a flat plate square to the axis returns the echo to both boards.

B Side elevation — heights, co-planar

The critical height rule: **the boards' antennas and the centre of the plate sit at the same height**, so the reflection stays in one horizontal plane and the echo comes straight back. A tripod or a stack of boxes works — just measure it.

Vertical arrangement

heights n.t.s. · keep coplanar

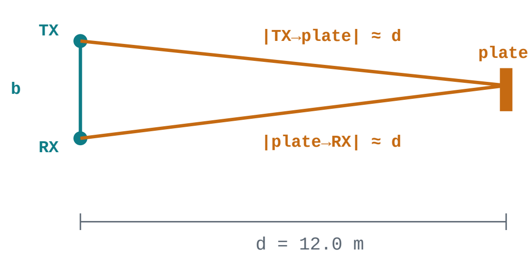


Exact height doesn't matter; matching the two heights does. ~1 m off the floor keeps the LOS and echo clear of floor/ceiling clutter.

C Where the numbers come from

Path lengths

b ≪ d



Echo path ≈ 2d. Excess over LOS = 2d − b.

The measured quantity

excess delay

c = **2.998×10⁸** m/s | B = **40** MHz

Excess path	2d − b = **23.5 m**
Excess delay	(2d−b)/c = **78 ns**
Path cell (c/B)	7.49 m
In cells	**3.13**
Range cell (c/2B)	**3.75 m**

This is the tap the differential should reveal.

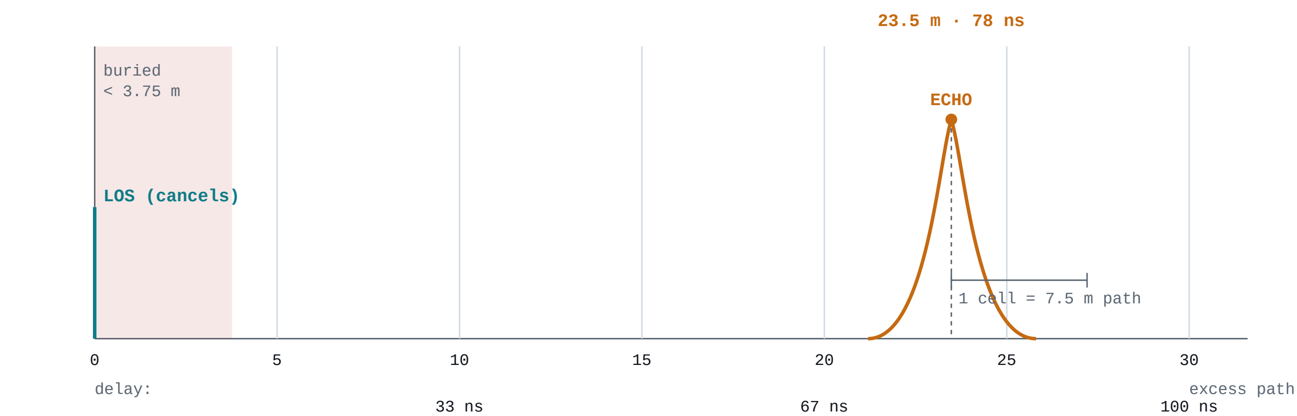
DRAWING 4 OF 4 — EXPECTED RESULT

D What you should see (differential delay profile)

After **plate-in minus plate-out**, the direct path cancels and a single tap remains at the echo's excess delay. Anything closer than one range cell (**`3.75 m`**) is swallowed by the direct path — which is why the plate starts at 12 m.

Differential |CIR| — prediction

x: excess path / delay



Move the plate 8 → 10 → 12 → 14 m; the tap must slide 6.67 ns per metre. No tap at any distance = the Rung-1 kill criterion (report the negative result).

BILL OF MATERIALS

E Elements & dimensions

ITEM	SPEC / DIMENSION	QTY	NOTES
ESP32-S3 board	DevKitC-1, HT40 CSI, on **COM/UART port**	2	TX + RX; ~$16
Metal plate	flat, conductive, ≥ 30 × 30 cm	1	baking tray; flat = specular
Tape measure	≥ 15 m (laser ± cm better)	1	the ground-truth instrument
Tripods / stands	hold boards + plate at one height	2	coplanar is the rule
USB cables + laptop	data-capable; venv installed	1	runs the capture + analysis

ITEM	SPEC / DIMENSION	QTY	NOTES
Corridor (site)	straight ≥ 15 m, width ≥ 2 m	1	the make-or-break requirement

`baseline b = 0.50 m` `plate d = 8 / 10 / 12 / 14 m` `height h = same for all`

`channel = 2.4 GHz #1` `records = 1000 / recording`

FINDING A LOCATION

F Site requirements — what to look for

- [ ] **Straight, clear ≥ 15 m.** A long hallway, warehouse aisle, empty parking lane, or gym.
- [ ] **Nothing within ~4 m** of the boards along the axis (closer than one range cell hides in the direct path).
- [ ] **Room to move the plate** to 8, 10, 12, 14 m and mark each spot.
- [ ] **Quiet during capture** — no people/carts crossing the path while recording (~10 s each).
- [ ] **Power** — a socket or a laptop battery; the boards run off USB.

Why not a small room? At 40 MHz the range cell is 3.75 m. In a 4–5 m room every wall echo lands in the same bin as the direct path and the differential shows nothing — you'd measure *nothing*, not a null result. The site size is physics, not preference.

ON-SITE PROCEDURE

G Setup & run — hit-run steps

01 **Place the boards** at the near end, 0.50 m apart, both facing down the corridor. Plug both into the laptop via their **COM/UART** ports. They power on and the TX starts transmitting automatically.

02 **Confirm the link (Rung 0.5).** Boards can be on a table for this. Run the check — it auto-finds the port and confirms one clean tap:

```
# from the repo root
.venv/bin/python experiments/run_bench.py check
```

03 **Set the geometry.** Boards 0.50 m apart; plate on the axis at 12 m, at the same height, facing the boards. Measure each with the tape.

04 **Run the atomic test.** It prompts you to record with the plate in, then to remove it and record out — twice — then prints the detected echo vs the prediction:

```
.venv/bin/python experiments/run_bench.py rung1 --plate 12
```

05 **Read the result.** A tap near **`23.5 m / 78 ns`** = success (first delay-resolved reflector from an ESP32). Then repeat at 10 and 14 m to confirm it tracks.

06 **If nothing appears** at 10, 12 *and* 14 m — that is the honest Rung-1 kill criterion. Save the captures; the negative result is itself a finding.

> **Captures auto-save** to `data/hw_captures/` with timestamps, so every run is reproducible offline with `experiments/run_hw_phantom.py`.
